Project Report on

"Predicting Impulse Buying Behavior using Psychological and Environmental Factors"

Submitted in Partial Fulfillment of the Requirements for the Award of degree of Post Graduate Diploma in Management 2025 – 2026

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Declaration

I, **<u>Harshit Saraf</u>**, hereby declare that the project report titled **<u>"Predicting Impulse Buying Behavior using Psychological and Environmental Factors"</u>** submitted to VES Business School, Mumbai, in partial fulfilment of the requirements for the award of the degree of Post Graduate Diploma in Management, is a result of my original work carried out under the guidance of **Prof Nikita Ramrakhiani.**

This report has not been submitted previously to any university or institute for the award of any degree, diploma, or certification.

Signature

Harshit Saraf

Date:

Acknowledgement

I express my gratitude to the staff members of VES Business School who directly or indirectly helped me. I express my sincere gratitude to my project guide, **Prof. Nikita Ramrakhiani**, for providing continuous support, guidance, and valuable insights throughout the course of this project.

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Signature

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1. Executive Summary

The rapid growth of e-commerce platforms has significantly transformed consumer purchasing behavior. With the increasing presence of flash sales, personalized recommendations, influencer marketing, and limited-time discounts, consumers are more frequently exposed to triggers that encourage spontaneous purchasing decisions. As a result, impulse buying has become a common phenomenon in online shopping environments. Understanding the factors that influence such behavior is important for businesses, marketers, and digital platforms that aim to optimize customer engagement and purchasing experiences.

The objective of this study is to analyze and predict impulse buying behavior by examining both psychological and environmental factors that influence consumer decisions during online shopping. Psychological factors include emotional states such as mood, stress, fear of missing out (FOMO), and personal self-control, while environmental factors involve external stimuli such as discounts, flash sales, notifications, influencer promotions, and peer recommendations. By studying these variables together, the research attempts to understand how internal emotions and external marketing triggers interact to influence purchasing behavior.

Primary data for this research was collected through a structured online questionnaire distributed to individuals who actively engage in online shopping. The questionnaire captured demographic information, recent purchase behavior, psychological tendencies, and exposure to digital marketing triggers. The collected responses were then cleaned, processed, and transformed into numerical variables suitable for analysis.

The study adopts a quantitative and predictive research approach by applying machine learning techniques to analyze the collected data. Multiple classification algorithms were implemented, including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting models. These models were trained to predict whether a purchase was impulsive or planned based on the collected behavioral and contextual variables. Model performance was evaluated using standard metrics such as accuracy, precision, recall, and F1-score.

The analysis identified several significant predictors of impulse buying behavior. Psychological factors such as fear of missing out, emotional influence during shopping, and difficulty resisting attractive offers showed strong relationships with impulsive purchasing decisions. Environmental triggers such as discounts, limited-time offers, and digital notifications were also found to increase the likelihood of impulse buying. Additionally, behavioral indicators like faster decision-making time and minimal product evaluation were associated with higher impulse buying tendencies.

Overall, the machine learning models demonstrated strong predictive capability in identifying impulse buying behavior from the collected data. The findings highlight the importance of both emotional and environmental triggers in shaping consumer purchase decisions. These insights can be valuable for e-commerce platforms, digital marketers, and product strategists seeking to understand customer behavior and design more effective marketing strategies.

In conclusion, this study demonstrates how machine learning and behavioral analytics can be applied to consumer research to uncover patterns in purchasing behavior. By combining psychological insights with predictive modeling, the research contributes to a better understanding of impulse buying in the modern digital marketplace.

2. Introduction

The growth of e-commerce and digital marketplaces has significantly changed the way consumers make purchasing decisions. Online shopping platforms provide easy access to a wide range of products, attractive discounts, personalized recommendations, and promotional offers that can strongly influence customer behavior. In such environments, many purchases are made spontaneously without prior planning, a phenomenon commonly known as impulse buying.

Impulse buying refers to unplanned purchases that occur when consumers make quick decisions driven by emotions, external triggers, or situational influences rather than careful evaluation. Factors such as fear of missing out (FOMO), limited-time offers, push notifications, influencer recommendations and discounts often encourage consumers to make instant purchasing decisions. As online platforms continue to adopt aggressive marketing strategies, understanding what drives impulse buying behavior has become increasingly important for businesses and marketers.

This study aims to analyze the key psychological and environmental factors that influence impulse buying behavior in online shopping. By collecting primary data through a survey and applying machine learning techniques, the research attempts to identify patterns that predict whether a purchase is impulsive or planned. The insights obtained from this analysis can help businesses better understand consumer behavior and develop more effective marketing strategies in the digital retail environment.

2.1 Background of the study

The increasing use of digital platforms and mobile technology has made online shopping more convenient and accessible for consumers. E-commerce platforms continuously introduce promotional strategies such as discounts, personalized recommendations, and targeted advertisements to attract customers and influence purchasing decisions. These strategies often encourage consumers to make quick decisions while browsing online stores.

At the same time, consumer decisions are not influenced only by marketing strategies but also by psychological factors such as emotions, excitement, and personal spending habits. The interaction between internal psychological tendencies and external marketing triggers can significantly affect whether a purchase is carefully planned or made impulsively.

Understanding these behavioral patterns has become important for businesses and researchers who aim to analyze consumer decision-making in digital environments. With the help of data-driven techniques such as machine learning, it is possible to identify patterns in purchasing behavior and predict the likelihood of impulse buying based on different influencing factors.

2.2 Need & Significance of the Study

With the rapid growth of online shopping, consumer purchasing behavior has become more dynamic and influenced by various digital marketing strategies. E-commerce platforms frequently use techniques such as discounts, flash sales, personalized recommendations, and notifications to attract customers and increase sales. These strategies often lead consumers to make unplanned or impulsive purchases. Therefore, understanding the factors that influence impulse buying behavior has become increasingly important for businesses and researchers.

This study is significant because it examines both psychological and environmental factors that affect consumer purchasing decisions in online shopping environments. By analyzing these factors using machine learning techniques, the study aims to identify patterns that can help predict impulse buying behavior.

The findings of this research can provide useful insights for e-commerce platforms, marketers, and businesses to better understand consumer behavior and design more effective marketing strategies. Additionally, the study demonstrates how data analytics and machine learning can be applied to consumer behavior research, contributing to the growing field of data-driven decision-making.

2.3 Objectives of the Study

- To identify the psychological factors that influence impulse buying behavior in online shopping.
- To examine the impact of environmental triggers such as discounts, flash sales, advertisements, and notifications on consumer purchase decisions.

- To analyze the relationship between consumer behavior patterns and impulse buying tendencies.
- To develop a machine learning model to predict whether a purchase is impulsive or planned based on collected data.
- To identify the most important factors that contribute to impulse buying behavior among online shoppers.

2.4 Scope of study

The scope of this study is to analyze impulse buying behavior among consumers who engage in online shopping. The research focuses on understanding how psychological factors such as emotions, stress and fear of missing out (FOMO), along with environmental triggers like discounts, flash sales, advertisements and notifications, influence purchasing decisions.

The study is based on primary data collected through a structured questionnaire from individuals who have experience with online shopping platforms. The analysis is limited to the responses collected during the survey period and focuses only on online purchase behavior.

Machine learning techniques are used to analyze the collected data and develop a predictive model that identifies patterns associated with impulse buying. The findings of the study aim to provide insights into consumer decision-making in digital shopping environments.

3. Statement of Problem

With the rapid growth of e-commerce platforms, consumers are increasingly exposed to various digital marketing strategies such as flash sales, discounts, personalized recommendations, and notifications. These strategies often encourage quick purchasing decisions, leading to a rise in impulse buying behavior. While such purchases may increase sales for businesses, they also reflect how consumer decisions are influenced by both emotional and environmental factors.

However, understanding which factors most strongly influence impulse buying remains a challenge. Consumers may be influenced by psychological triggers such as mood, excitement, or fear of missing out (FOMO), as well as external stimuli like promotional offers and advertisements. Identifying the key drivers behind these spontaneous purchasing decisions is important for businesses seeking to better understand consumer behavior.

Therefore, this study attempts to examine the factors influencing impulse buying in online shopping and develop a machine learning model that can predict whether a purchase is impulsive or planned based on consumer responses.

4. Literature Review

1. Influence of FOMO on Shopping Motivation and Compulsive Buying - Barbu Kleitsch & Drămnescu (2025)

Barbu Kleitsch and Drămnescu (2025) explored how Fear of Missing Out (FOMO) affects shopping motivation and compulsive buying behavior among young adults. The study is grounded in the idea that digital environments, especially social media and online retail spaces, constantly expose users to comparisons, trends, promotions, and urgency-based cues. These conditions make young consumers more vulnerable to the feeling that they may miss an important opportunity, product, or social experience if they do not act immediately. The authors argue that this emotional pressure can shift shopping from a rational activity to a more reactive and emotionally driven one.

The article highlights that FOMO is not limited to social interaction; it also extends into consumer behavior, particularly in situations where products are presented as limited, trending, or socially valued. In such settings, consumers may feel a psychological urge to buy quickly before the opportunity disappears. This tendency is especially visible among young adults, who are more engaged with digital content, online communities, and social comparison processes. The study suggests that FOMO increases shopping motivation by creating a sense of urgency and emotional excitement, which can then lead to compulsive or uncontrolled purchasing behavior.

A major contribution of this article is its explanation of the connection between psychological states and purchase behavior. Rather than seeing buying decisions only as responses to price or product need, the study emphasizes internal emotional triggers. It shows that consumers experiencing stronger FOMO are more likely to make sudden purchases with less deliberation. Over time, these repeated impulsive decisions may even develop into compulsive buying patterns. This is important because it demonstrates that emotional variables can play a strong predictive role in online shopping behavior.

For the present study, this article is highly relevant because it directly supports the inclusion of FOMO as a key psychological variable in predicting impulse buying behavior. Your project examines whether psychological and environmental factors influence impulsive online purchases, and this paper gives a strong theoretical basis for using FOMO-related survey items in the model. It also strengthens the argument that digital shopping platforms do not simply provide products; they actively create emotional conditions that can shape purchasing decisions. Overall, this study is useful in establishing that emotional urgency and fear of missing out are important drivers of unplanned buying behavior in digital environments.

Source: https://www.mdpi.com/2673-5172/6/3/139?utm_source

2. Factors Influencing Online Impulse Buying Behavior on Shopee Video Platform - Ngo et al. (2024)

Ngo et al. (2024) conducted a comprehensive study on online impulse buying behavior using evidence from the Shopee video platform, a digital commerce environment where products are promoted through interactive and short-form video content. The study focuses on how different online stimuli influence consumer emotions and lead to unplanned purchase behavior. Unlike traditional online shopping studies that focus mainly on website design or discounts, this research pays attention to the role of dynamic digital content and platform-specific shopping experiences.

The study identifies several key factors that influence online impulse buying, including time pressure, quantity pressure, economic benefits, social influence, and entertainment value. These factors act as external stimuli in the online shopping environment. For example, limited-time offers and low-stock messages create urgency, while price benefits and promotional content increase the attractiveness of the purchase. Social influence also plays an important role, especially when users are exposed to content that suggests popularity, recommendation, or approval from others. According to the study, these platform-driven factors stimulate emotional responses such as excitement, enjoyment, and urge to buy, which then increase the likelihood of impulse purchasing.

An important insight from the article is that online impulse buying is not caused by a single factor, but by the interaction of multiple environmental and emotional variables. The platform's content presentation, combined with marketing triggers, shapes how consumers feel during the shopping process. When individuals experience pleasure, urgency, or social influence while browsing, they become more likely to make quick and unplanned decisions. This finding is valuable because it shows that impulse buying behavior can be studied through a structured, data-driven framework where external stimuli and internal emotional reactions are linked.

This article is highly relevant to your project because it provides direct support for including environmental variables such as discounts, limited-time offers, peer influence, advertisements, and promotional exposure in your predictive model. It also reinforces the idea that online platforms are designed in ways that can trigger spontaneous decisions. For your capstone, this study helps justify why environmental factors should be analyzed alongside psychological ones. In short, the paper shows that digital shopping behavior is strongly affected by platform-based cues, and these cues can be powerful predictors of impulse buying when modeled systematically.

Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC11336989/?utm_source

5. Research Methodology

This study adopts a quantitative, descriptive, and predictive research approach to examine impulse buying behavior in online shopping environments. The research is based on primary data collected through a structured questionnaire designed using Google Forms. The survey targeted individuals who actively engage in online shopping and included questions related to demographic details, recent purchase behavior, psychological factors (such as mood, stress, and fear of missing out) and environmental triggers (such as discounts, flash sales, notifications, and influencer influence).

A convenience sampling technique was used to gather responses, and a total of **295 valid responses** were collected for analysis. The collected data was then prepared for analysis by handling missing values, encoding categorical variables into numerical format, and creating additional features such as psychological and environmental scores. The target variable was defined as whether a purchase was impulsive or planned.

Data analysis was carried out using Python, with the help of libraries such as Pandas, NumPy, and Scikit-learn. Initially, Exploratory Data Analysis (EDA) was performed to understand the distribution of variables and identify patterns and relationships in the data. Subsequently, multiple machine learning classification models were applied, including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting, to predict impulse buying behavior.

The performance of these models was evaluated using metrics such as accuracy, precision, recall, and F1-score, along with confusion matrix analysis. Additionally, feature importance techniques were used to identify the most significant factors influencing impulse buying behavior.

This methodology provides a systematic and data-driven approach to understanding consumer behavior and developing a reliable predictive model for impulse buying in online shopping environments.

5.1 Research Design

This study follows a quantitative and predictive research design aimed at analyzing impulse buying behavior in online shopping environments. The design focuses on collecting structured numerical data and applying analytical techniques to identify patterns and relationships between variables.

The research is descriptive in nature, as it seeks to describe consumer behavior and understand how different psychological and environmental factors influence purchasing decisions. At the same time, it is predictive, as machine learning models are used to predict whether a purchase is impulsive or planned based on the collected data.

A cross-sectional research design is adopted, where data is collected from respondents at a single point in time through a structured questionnaire. The design ensures that various factors such as emotions, external triggers, and behavioral patterns are captured simultaneously and analyzed collectively.

Overall, this research design enables a systematic examination of consumer behavior and supports the development of a data-driven model to predict impulse buying in digital shopping environments.

5.2 Sampling Method & Sample Size

The study uses a non-probability convenience sampling method, where respondents were selected based on their accessibility and willingness to participate in the survey. The questionnaire was distributed through online platforms such as WhatsApp and other social networks, targeting individuals who actively engage in online shopping.

A total of **295** valid responses were collected and used for analysis. The sample includes respondents from different age groups, occupations, and income levels to capture diverse consumer behavior patterns.

The chosen sampling method is suitable for this study due to time constraints and ease of data collection, while still providing sufficient data to perform statistical analysis and build machine learning models.

5.3 Questionnaire Design / Variables

The questionnaire for this study was designed in a structured format to collect data related to consumer behavior in online shopping environments. It consisted of close-ended questions to ensure ease of response and to facilitate quantitative analysis. The questions were divided into different sections to capture various aspects influencing impulse buying behavior. (Annexure A)

- **Demographic Variables**

These variables were included to understand the background of respondents and to analyze behavioral differences across groups.

- Age
- Gender
- Occupation
- Which platform did you use for your recent purchase?
- Device used during purchase
- How often do you shop Online?

- **Purchase Behavior Variables**

These variables capture the context of the respondent's recent purchase.

- Think about your most recent online purchase. Was it planned in advance?
- Did you originally intend to buy this product before opening the app/website?
- How much time did you spend browsing before purchasing?

- **Psychological Variables**

These variables measure internal emotional and behavioral tendencies of consumers using Likert scale (1–5).

- I often buy things when I am feeling stressed.
- I shop online to improve my mood.
- I find it hard to resist attractive offers.
- I experience FOMO (Fear of Missing Out) during flash sales.
- I sometimes buy products without thinking about future need.
- I feel excited while making spontaneous (Random) purchases.
- I regret some purchases later.
- I consider myself emotionally influenced while shopping.

- **Environmental Variables**

These variables represent external triggers that may influence purchasing decisions.

- Was the product on discount?
- Did you see a “Limited Time Offer” message?
- Did an influencer/advertisement influence this purchase?
- Did you receive a push notification before buying?
- Discount percentage (if applicable)
- Did peer (Known Person) recommendation influence your purchase?
- Payment Method Used (Mostly)
- Do you usually compare prices before purchasing?
- How often do you add items to cart but do not purchase?
- On average, how many impulse (Actual) purchases do you make per month?

- **Behavioral Variables**

These variables reflect rational decision-making patterns of consumers.

- I carefully plan my purchases in advance.
- I stick to my budget while shopping.
- I am comfortable spending money without long consideration.
- I review my expenses regularly.

- **Target Variable** (Annexure B)

The dependent variable in this study is Impulse Buying Behavior, which classifies whether a purchase was:

- Planned
- Partially planned
- Completely impulsive

For analysis, this variable is converted into a binary format:

1 = Impulsive Purchase

0 = Planned Purchase

5.4 Data Collection Method

The data for this study was collected using a primary data collection method through a structured questionnaire. The questionnaire was designed using Google Forms and distributed online to respondents who actively engage in online shopping.

The survey link was shared through various digital platforms such as WhatsApp and other social networks, making it easily accessible to a wide range of participants. Respondents were requested to answer the questions based on their recent online shopping experience, ensuring that the responses reflected actual behavior.

The questionnaire consisted of close-ended questions, including multiple-choice and Likert scale items, to facilitate ease of response and enable quantitative analysis. All responses were collected anonymously to ensure honesty and reduce response bias.

The collected data was automatically recorded in a spreadsheet format, which was later exported for data cleaning, processing, and analysis using analytical tools. (Annexure C)

5.5 Tools & Techniques for Analysis

Tools Used:

- **Python** was used as the primary tool for data analysis and model development.
- **Google Colab** was used as the working environment for implementation.
- **Pandas and NumPy** libraries were used for data cleaning, manipulation, and preparation.
- **Matplotlib and Seaborn** were used for data visualization and exploratory analysis.
- **Scikit-learn** was used for building and evaluating machine learning models.

Techniques Used:

1. Data Preprocessing

The collected data was cleaned and prepared by handling missing values, encoding categorical variables, and transforming the data into a suitable format for analysis.

2. Exploratory Data Analysis (EDA)

EDA was performed to understand the distribution of variables, identify patterns, and examine relationships between different factors influencing impulse buying behavior.

3. Machine Learning Models

Classification techniques were applied to predict whether a purchase was impulsive or planned. The models used include:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

4. Model Evaluation

The performance of the models was evaluated using standard metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

5. Feature Importance Analysis

Feature importance techniques were used to identify the most influential psychological and environmental factors affecting impulse buying behavior.

6. Data Analysis & Interpretation

- **Overview:**

The dataset for this study was collected through a structured online questionnaire using Google Forms. A total of 295 valid responses were obtained from individuals who actively engage in online shopping. The dataset includes multiple variables covering demographic information, purchase behavior, psychological factors, environmental triggers, and behavioral patterns.

The target variable for the study is Impulse Buying Behavior, which classifies whether a purchase was impulsive or planned. For analytical purposes, this variable was converted into a binary format, where 1 represents an impulsive purchase and 0 represents a planned purchase.

- **Data Preprocessing and Preparation:**

Before analysis, the collected data was cleaned and prepared for modeling. Missing values were handled appropriately and categorical variables such as gender, platform and device were encoded into numerical format. Additionally, new features such as psychological score and environmental trigger score were created by aggregating relevant variables.

The dataset was then divided into training and testing sets using an 80:20 split to evaluate model performance effectively. (Annexure D & E)

- **Exploratory Data Analysis (EDA)**

3.1 Distribution of Impulse Buying Behavior

The analysis of the target variable shows that a considerable proportion of respondents reported making impulsive purchases. This indicates that impulse buying is a prevalent behavior in online shopping environments.

Interpretation:

The high occurrence of impulse buying suggests that digital platforms and marketing strategies significantly influence consumer decision-making.

3.2 Analysis of Psychological Factors

Psychological variables such as FOMO, emotional influence, stress-based shopping, and excitement during spontaneous purchases showed higher average values among respondents who made impulsive purchases.

Interpretation:

Consumers who experience stronger emotional triggers are more likely to engage in impulse buying. Fear of missing out (FOMO) and emotional excitement play a critical role in reducing rational decision-making.

3.3 Analysis of Environmental Factors

Environmental variables such as discounts, limited-time offers, push notifications, and influencer exposure were strongly associated with impulsive purchases.

Interpretation:

External marketing strategies create urgency and attractiveness, which encourages consumers to make quick purchase decisions without thorough evaluation.

3.4 Behavioral Pattern Analysis

Respondents who exhibited the following behaviors were more likely to make impulsive purchases:

Spending less time evaluating products

Making faster purchase decisions

Not comparing prices

Skipping product reviews

Interpretation:

Lower cognitive effort and quicker decision-making are key indicators of impulse buying behavior.

- **Machine Learning Model Analysis:** (Annexure F)

To predict impulse buying behavior, multiple classification models were applied, including:

Logistic Regression

Decision Tree

Random Forest

Gradient Boosting

Each model was trained using the prepared dataset and evaluated on the test data.

Among these, **Random Forest** demonstrated the best performance due to its ability to handle complex relationships and interactions between variables.

- **Model Performance Evaluation** (Annexure G)

The models were evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score.

The best-performing model achieved an accuracy of approximately **73 %**, indicating a strong ability to correctly classify impulsive and planned purchases.

Interpretation:

The high accuracy of the model suggests that the selected variables effectively capture the patterns associated with impulse buying behavior.

- **Feature Importance Analysis** (Annexure H & J)

Feature importance analysis was conducted to identify the most influential factors affecting impulse buying behavior. The top contributing variables include:

- Fear of Missing Out (FOMO)
- Emotional influence during shopping
- Difficulty in resisting attractive offers
- Discounts and promotional offers
- Decision-making speed

Interpretation:

Both psychological and environmental factors significantly influence impulse buying behavior. Emotional triggers combined with external marketing cues increase the likelihood of spontaneous purchases.

- **Impulse Risk Segmentation**

Based on the model predictions, respondents were segmented into different categories:

Low Impulse Buyers – Individuals who mostly make planned purchases

Moderate Impulse Buyers – Individuals who occasionally make impulsive purchases

High Impulse Buyers – Individuals who frequently engage in impulse buying

Interpretation:

This segmentation helps in understanding different consumer groups and can be useful for businesses to design targeted marketing strategies.

- **Overall Interpretation**

The overall analysis indicates that impulse buying behavior is influenced by a combination of internal psychological factors and external environmental triggers. Emotional states such as excitement and FOMO, along with marketing strategies like discounts and limited-time offers, significantly increase the likelihood of impulsive purchases.

The machine learning model successfully captures these relationships and provides a predictive framework for identifying impulse buying behavior. This demonstrates the effectiveness of combining behavioral analysis with data-driven techniques in understanding consumer decision-making.

7. Findings / Results & Discussion

Key Findings:

Based on the analysis of the collected data and the application of machine learning models, the following key findings were observed:

- **High Prevalence of Impulse Buying**

A significant proportion of respondents reported making unplanned or impulsive purchases during online shopping.

Discussion:

This indicates that impulse buying is a common behavior in digital commerce, largely influenced by the design and marketing strategies of online platforms.

- **Psychological Factors Strongly Influence Behaviour**

Variables such as Fear of Missing Out (FOMO), emotional influence, excitement, and difficulty in resisting offers were found to be strong predictors of impulse buying.

Discussion:

Consumers experiencing higher emotional engagement are more likely to make spontaneous purchase decisions. This supports the idea that impulse buying is driven more by emotions than rational thinking.

- **Environmental Triggers Play a Major Role**

External factors such as discounts, limited-time offers, push notifications, and advertisements were strongly associated with impulsive purchases.

Discussion:

Marketing strategies that create urgency and perceived value significantly increase the likelihood of impulse buying. These triggers reduce the time available for rational decision-making.

- **Faster Decision-Making Leads to Impulse Buying**

Respondents who made quicker purchase decisions and spent less time evaluating products were more likely to exhibit impulse buying behavior.

Discussion:

Impulse buying is characterized by low cognitive effort and quick decision-making, where consumers do not thoroughly analyze alternatives.

- **Machine Learning Model Effectiveness**

Among the applied models, Random Forest performed the best, achieving an accuracy of approximately 73 %.

Discussion:

The model's performance indicates that the selected variables effectively capture patterns associated with impulse buying. It also demonstrates the usefulness of machine learning in predicting consumer behavior.

- ## **Most Influential Factors Identified**

Feature importance analysis revealed that the most significant factors influencing impulse buying include:

- FOMO (Fear of Missing Out)
- Emotional influence during shopping
- Discounts and promotional offers
- Difficulty in resisting attractive offers
- Decision-making speed

Discussion:

These findings confirm that both internal psychological factors and external marketing triggers jointly influence consumer decisions.

- ## **Consumer Segmentation Based on Impulse Behaviour**

The analysis enabled segmentation of consumers into:

Low impulse buyers

Moderate impulse buyers

High impulse buyers

Discussion:

This segmentation can help businesses design targeted marketing strategies and personalize user experiences based on consumer behavior patterns.

- ## Overall Discussion

The results of this study clearly demonstrate that impulse buying behavior in online shopping is influenced by a combination of emotional, behavioral, and environmental factors. Psychological triggers such as FOMO and emotional excitement increase the likelihood of impulsive decisions, while external factors such as discounts and promotional messages act as catalysts.

The findings also highlight that modern e-commerce platforms are designed in a way that encourages quick decision-making, often reducing the role of rational evaluation. The successful application of machine learning models further shows that impulse buying behavior can be predicted using structured data, making it valuable for businesses aiming to optimize marketing strategies and customer engagement.

8. Learnings / Managerial Implications

This study provides important insights into consumer behavior in online shopping environments, particularly in relation to impulse buying. The findings indicate that impulse buying is influenced by a combination of psychological factors, environmental triggers, and behavioral patterns. Emotional factors such as fear of missing out (FOMO), excitement, and emotional influence during shopping play a significant role in shaping consumer decisions. At the same time, external factors such as discounts, limited-time offers, notifications, and advertisements act as strong triggers that encourage unplanned purchases.

The study also highlights that consumers who make faster decisions and spend less time evaluating alternatives are more likely to engage in impulse buying. This suggests that reduced cognitive effort and quick decision-making are key characteristics of impulsive purchasing behavior. Additionally, the application of machine learning techniques in this study demonstrates that consumer behavior can be effectively analyzed and predicted using data-driven approaches.

From a managerial perspective, the findings of this study have several practical implications. Businesses can design more effective marketing strategies by leveraging psychological triggers and environmental cues such as flash sales, personalized recommendations, and promotional offers. These strategies can help increase customer engagement and drive sales.

Furthermore, the segmentation of consumers based on impulse buying behavior enables organizations to implement targeted marketing approaches. For instance, customers who exhibit higher impulsive tendencies can be targeted with promotional campaigns, while more rational consumers may respond better to detailed product information and comparisons.

The study also emphasizes the importance of using behavioral data for decision-making. Organizations can analyze customer interactions and purchasing patterns to improve customer experience, enhance retention, and develop personalized marketing strategies.

However, it is important for businesses to consider ethical aspects while applying such strategies. Excessive use of psychological triggers may lead to post-purchase regret and negatively impact customer satisfaction and long-term loyalty.

Overall, the study demonstrates that integrating behavioral insights with data analytics can help organizations better understand consumer decision-making and develop effective, data-driven marketing strategies in the digital marketplace.

9. Limitations of the Study

- **Limited Sample Size**

  The study is based on a relatively limited number of responses, which may not fully represent the entire population of online shoppers. A larger sample size could provide more accurate and generalizable results.

- **Convenience Sampling Method**

  The study uses a convenience sampling technique, where respondents were selected based on accessibility. This may introduce sampling bias, as the sample may not reflect the characteristics of the broader population.

- **Self-Reported Data**

  The data collected is based on respondents' self-reported answers, which may be subject to bias. Respondents may not always accurately recall their behavior or may provide socially desirable responses instead of their actual actions.

- **Focus on Online Shopping Only**

  The study is limited to online shopping behavior and does not consider impulse buying in offline retail environments. Therefore, the findings may not be applicable to traditional shopping contexts.

- **Cross-Sectional Nature of Study**

  The data was collected at a single point in time. Consumer behavior may change over time due to external factors such as trends, economic conditions, or personal circumstances, which are not captured in this study.

- **Limited Variables Considered**

  Although the study includes key psychological and environmental factors, there may be other variables such as cultural influences, personality traits, or brand loyalty that were not included but could also affect impulse buying behavior.

- **Model Generalization Limitations**

  The machine learning model is trained on the collected dataset and may not perform equally well on different datasets or populations. Its predictive accuracy is limited to the scope of the data used in this study.

- **Simplification of Target Variable**

  Impulse buying behaviour was converted into a binary variable (impulsive vs planned), which may oversimplify the complexity of consumer decision-making. In reality, purchasing behaviour exists on a spectrum.

- **Potential Response Bias**

  Since the survey was distributed online, it may have attracted respondents who are more active on digital platforms, leading to possible bias in the data.

10. Future Scope

The present study provides a comprehensive understanding of impulse buying behavior in online shopping environments by integrating psychological, environmental, and behavioral factors with machine learning techniques. While the study offers valuable insights, it also opens several avenues for future research that can further enhance the depth, accuracy, and applicability of the findings.

Firstly, future research can expand the scope of the study by increasing the sample size and including a more diverse group of respondents. A larger and more representative sample covering different age groups, income levels, professions, and geographical regions would improve the generalizability of the results. It would also allow researchers to examine how impulse buying behavior varies across different demographic segments and cultural contexts, thereby providing more nuanced insights into consumer behavior.

Secondly, the current study is limited to online shopping environments. Future studies can extend the research by incorporating offline retail settings and conducting a comparative analysis between online and offline impulse buying behavior. This would help in understanding how different environmental factors such as store layout, product placement, and salesperson interaction influence impulsive purchasing in physical stores, in contrast to digital triggers like notifications and flash sales.

Furthermore, additional variables can be incorporated to enrich the analysis. Factors such as personality traits, brand loyalty, lifestyle, financial awareness, and long-term consumption habits can provide a deeper understanding of consumer decision-making processes. Psychological constructs such as impulsivity, self-control, and risk-taking behavior can also be measured more extensively using validated scales to improve the reliability of the findings.

From a methodological perspective, future research can explore more advanced analytical techniques to enhance predictive performance. While this study uses traditional machine learning models, future studies can apply advanced models such as deep learning, neural networks, or ensemble techniques to capture more complex patterns in consumer behavior. In addition, time-series analysis can be used to study how impulse buying behavior evolves over time, especially during events such as festive sales or promotional campaigns.

Another important area for future research is the integration of real-time and behavioral data from e-commerce platforms. Instead of relying solely on self-reported survey data, researchers can utilize clickstream data, browsing history, purchase frequency, and session duration to build more accurate and dynamic predictive models. This would significantly enhance the practical applicability of the research in real-world business settings.

Moreover, future studies can focus on developing practical applications such as predictive dashboards, recommendation systems, or decision-support tools for businesses. These tools can help organizations identify potential impulse buyers, optimize marketing strategies, and personalize user experiences. Such applications can bridge the gap between academic research and industry implementation.

In addition, there is a need to explore the ethical implications of using psychological and behavioral insights in marketing. While impulse buying strategies can increase sales, excessive use of such techniques may lead to consumer dissatisfaction, financial stress, or post-purchase regret. Future research can examine how businesses can balance profitability with ethical responsibility and long-term customer well-being.

Finally, future research can adopt a longitudinal approach to study changes in consumer behavior over time. This would help in understanding how external factors such as economic conditions, technological advancements, and changing consumer preferences influence impulse buying patterns in the long run.

In conclusion, the future scope of this study lies in expanding the dataset, incorporating additional behavioral and psychological variables, applying advanced analytical techniques, and developing real-world applications. Such advancements can contribute to a more comprehensive and actionable understanding of consumer behavior in the evolving digital marketplace.

11. Conclusion & Recommendations

Conclusion:

The present study aimed to analyze and predict impulse buying behavior in online shopping environments by examining the influence of psychological and environmental factors. With the rapid growth of e-commerce and digital marketing strategies, consumer purchasing behavior has become increasingly dynamic and influenced by both internal emotions and external stimuli.

The findings of the study indicate that impulse buying is a common phenomenon among online shoppers. Psychological factors such as Fear of Missing Out (FOMO), emotional influence, excitement, and difficulty in resisting attractive offers were found to significantly affect purchasing decisions. At the same time, environmental triggers such as discounts, limited-time offers, notifications, and advertisements play a crucial role in encouraging unplanned purchases.

The application of machine learning techniques further strengthened the analysis by enabling the prediction of impulse buying behavior based on multiple variables. The results demonstrate that a combination of psychological and environmental factors can effectively explain and predict consumer behavior in online shopping contexts.

Overall, the study highlights that impulse buying is not random but is driven by identifiable patterns that can be analyzed and modeled. The integration of behavioral insights with data analytics provides a valuable approach for understanding consumer decision-making in the digital marketplace.

Recommendations

Based on the findings of the study, the following recommendations are suggested for businesses and e-commerce platforms:

- **Use of Targeted Marketing Strategies**

Organizations should design marketing campaigns that effectively utilize psychological triggers such as urgency, scarcity, and personalization. Techniques such as flash sales, limited-time offers, and personalized recommendations can significantly influence purchasing behavior.

- **Customer Segmentation and Personalization**

Businesses should segment customers based on their impulse buying tendencies and tailor marketing strategies accordingly. High impulse buyers can be targeted with promotional offers, while more rational consumers may benefit from detailed product information and comparisons.

- **Optimization of User Experience**

E-commerce platforms should focus on enhancing user experience by strategically placing product recommendations, discounts, and notifications. A well-designed interface can guide users toward purchase decisions while maintaining ease of navigation.

- **Leveraging Data Analytics for Decision-Making**

Organizations should utilize data analytics and machine learning techniques to analyze customer behavior and predict purchasing patterns. This can help in improving customer engagement, increasing conversion rates, and optimizing marketing strategies.

- **Balanced Use of Marketing Triggers**

While promotional strategies are effective, businesses should avoid excessive use of psychological triggers that may lead to customer dissatisfaction or regret. Maintaining a balance is important for long-term customer trust and loyalty.

- **Focus on Long-Term Customer Relationships**

Instead of focusing solely on short-term sales through impulse buying, businesses should also prioritize building long-term relationships with customers by ensuring satisfaction, transparency, and value delivery.

Overall Summary

In conclusion, this study demonstrates that impulse buying behavior in online shopping can be effectively understood and predicted using a combination of psychological insights and data-driven techniques. The findings provide valuable implications for businesses to develop smarter, more effective, and customer-centric strategies in the evolving digital economy.

12. References

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- https://jmsr-online.com/article/understanding-the-psychology-of-impulse-buying-in-e-commerce-a-behavioral-review-314/
- https://www.sciencedirect.com/science/article/abs/pii/S096969892300259X
- https://indianjournalofcapitalmarkets.com/index.php/ijom/article/view/175451

13. Annexures

Annexure (A)

```
============================================================
  STEP 1: Loading Data
============================================================
  Loaded 296 responses with 33 columns

  Columns found:
    - Timestamp
    - Age
    - Gender
    - Occupation
    - Which platform did you use for your recent purchase?
    - Device used during purchase
    - How often do you shop Online ?
    - Think about your most recent online purchase. Was it planned in advance?
    -   Did you originally intend to buy this product before opening the app/website?
    - How much time did you spend browsing before purchasing?
    - I often buy things when I am feeling stressed.
    - I shop online to improve my mood.
    -  I find it hard to resist attractive offers.
    - I experience FOMO (Fear of Missing Out) during flash sales.
    - I sometimes buy products without thinking about future need.
    - I feel excited while making spontaneous (Random)  purchases.
    - I regret some purchases later.
    - I consider myself emotionally influenced while shopping.
    - Was the product on discount?
    - Did you see a "Limited Time Offer" message?
    - Did an influencer/advertisement influence this purchase?
    - Did you receive a push notification before buying?
    - Discount percentage (if applicable)
    - Did peer (Known Person) recommendation influence your purchase?
    - Payment Method Used (Mostly)
    - Do you usually compare prices before purchasing?
```

Annexure (B)

```
============================================================
  STEP 3: Creating Target Variable
============================================================
  Target distribution:
    Impulse buyers  (1): 213 (72.0%)
    Planned buyers  (0): 83 (28.0%)
```

Annxure (C)

```
df.head()
```

age	gender	occupation	platform	device	shop_frequency	purchase_planned	intended_before_opening	browsing_time	stress_buying	...	peer_recommendation	payment_method
21 - 24	Male	Student	Flipkart	Mobile Browser	2 - 3 times a month	Yes	Yes	Less than 5 minutes	3	...	Yes	UPI (Gpay, Phonepe, Paytm)
21 - 24	Male	Student	Amazon	Mobile App	Rarely	Yes	Yes	Less than 5 minutes	1	...	No	UPI (Gpay, Phonepe, Paytm)
21 - 24	Female	Student	Amazon	Mobile App	Rarely	No	Yes	15 - 30 minutes	3	...	No	UPI (Gpay, Phonepe, Paytm)
21 - 24	Male	Student	Amazon	Mobile App	Rarely	Yes	Yes	15 - 30 minutes	1	...	No	UPI (Gpay, Phonepe, Paytm)
21 - 24	Male	Working Professional	Ajio	Mobile App	2 - 3 times a month	Yes	Yes	15 - 30 minutes	2	...	No	UPI (Gpay, Phonepe, Paytm)

ws × 32 columns

Annexure (D)

```
FEATURES = [
    # Demographics
    "age", "gender", "occupation",
    # Shopping behavior
    "platform", "device", "shop_frequency",
    # Purchase context
    "purchase_planned", "intended_before_opening", "browsing_time",
    # Psychological factors
    "stress_buying", "mood_shopping", "hard_resist_offers",
    "fomo_flash_sales", "buy_without_need", "excited_spontaneous",
    "regret_purchases", "emotionally_influenced",
    # Environmental triggers
    "product_on_discount", "limited_time_offer", "influencer_influenced",
    "push_notification", "discount_percentage", "peer_recommendation",
    # Payment & behavior
    "payment_method", "compare_prices", "add_cart_no_buy", "impulse_count_monthly",
    # Decision behavior
    "plan_purchases", "stick_budget", "comfortable_spending", "review_expenses",
    # Engineered features
    "psych_impulse_score", "env_trigger_score", "self_control_score", "impulse_risk_score"
]

TARGET = "is_impulse"

X = df[FEATURES]
y = df[TARGET]
```

Annexure (E)

```
print("\n" + "="*60)
print("  STEP 6: Train/Test Split")
print("="*60)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled  = scaler.transform(X_test)

print(f"  Training set : {len(X_train)} samples")
print(f"  Test set     : {len(X_test)} samples")
```

```
============================================================
  STEP 6: Train/Test Split
============================================================
  Training set : 236 samples
  Test set     : 60 samples
```

Annexure (F)

```
==================================================
  STEP 7: Training Models
==================================================

  Logistic Regression:
    Accuracy     : 70.00%
    ROC-AUC      : 0.7688
    CV Accuracy  : 74.57%

  Decision Tree:
    Accuracy     : 66.67%
    ROC-AUC      : 0.6915
    CV Accuracy  : 72.04%

  Random Forest:
    Accuracy     : 73.33%
    ROC-AUC      : 0.8222
    CV Accuracy  : 72.88%

  Gradient Boosting:
    Accuracy     : 75.00%
    ROC-AUC      : 0.7880
    CV Accuracy  : 71.21%
```

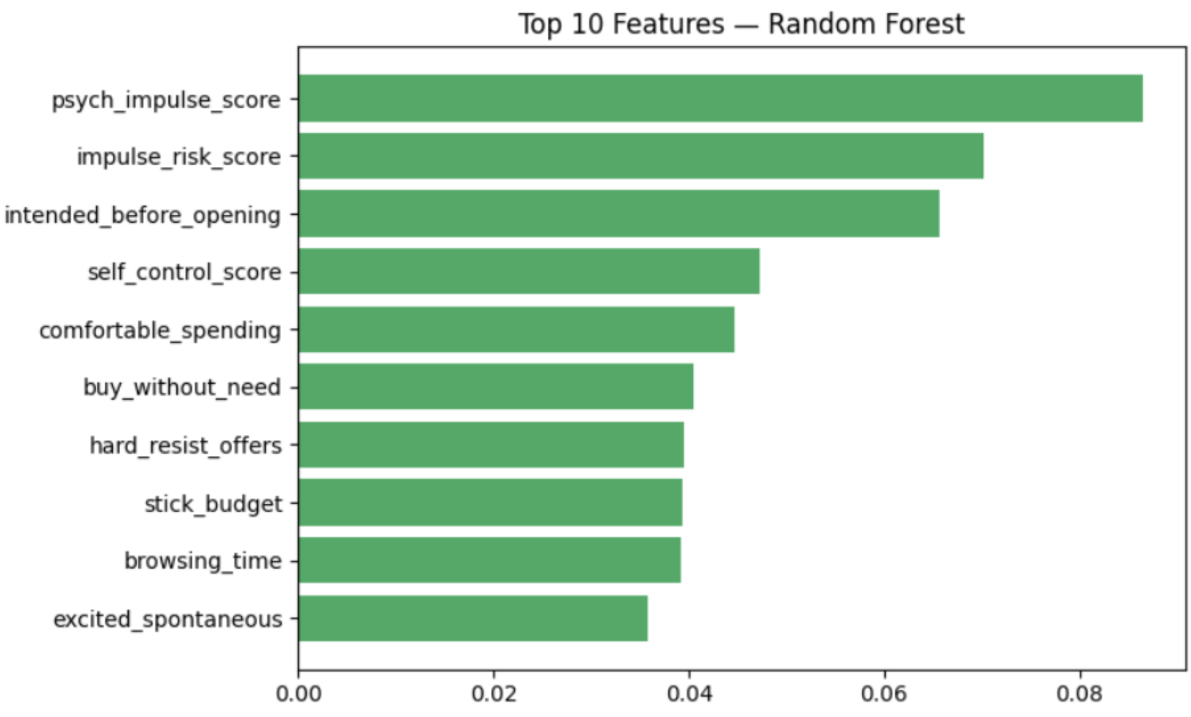
Annexure (G)

— Full Model Scorecard ————————————————

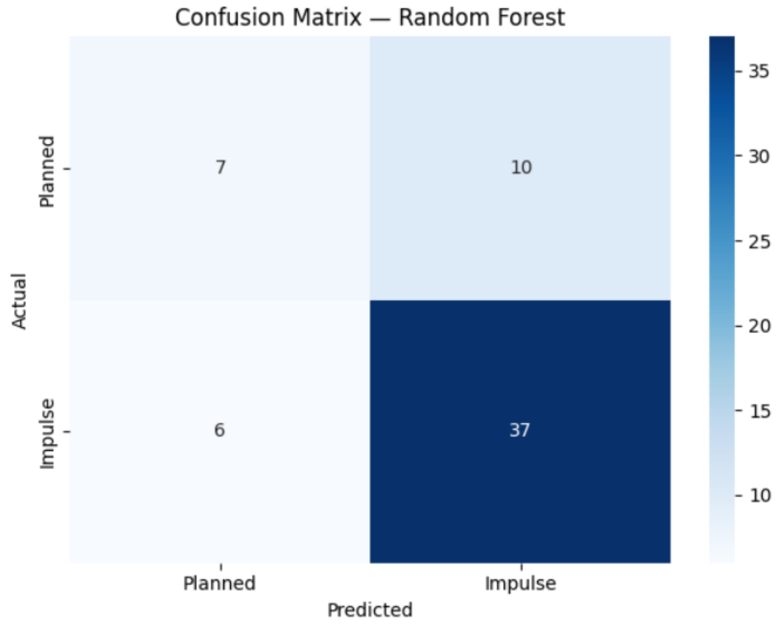
Model	Accuracy	AUC	CV Acc	Score
Logistic Regression	70.0%	0.769	74.6%	0.740
Decision Tree	66.7%	0.692	72.0%	0.700
Random Forest	73.3%	0.822	72.9%	0.753
Gradient Boosting	75.0%	0.788	71.2%	0.741

★ Best Model: Random Forest ★

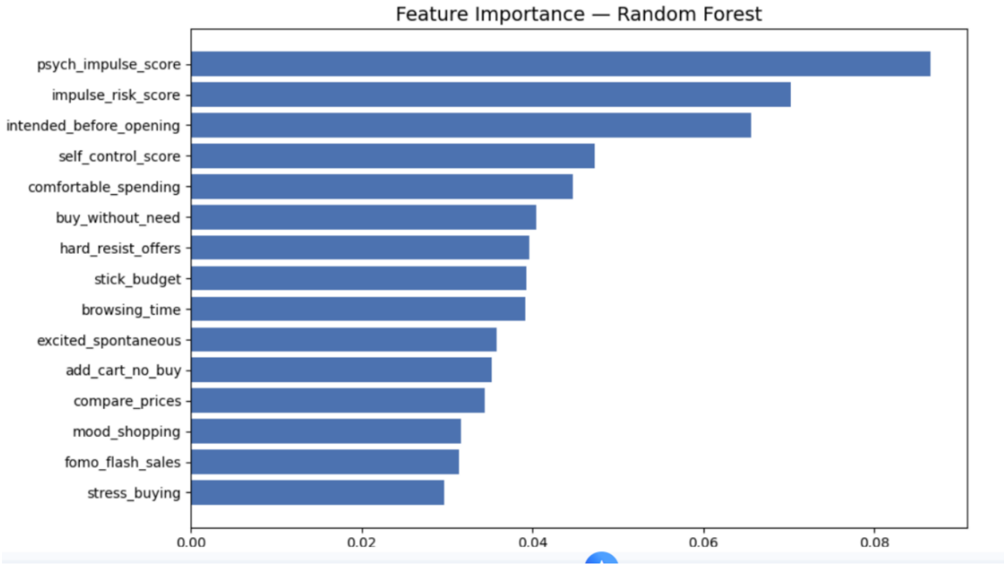
Annexure (H)



Annexure (I)



Annexure (J)



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Details Name of Mentee: **Mr. Harshit Saraf**

Sr. No.	Dates	Topic of discussion during the meeting	Faculty Signature

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